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一、全文中英文主旨

中文主旨

本文围绕科学领域的**专业化与职业化**两大发展趋势展开，开篇点明专业化是应对科学知识持续积累的产物，同时指出职业化是科学领域另一项重要变革；随后分析了19世纪专业化发展抬高了科研领域业余从业者的参与门槛，二者之间并无绝对界限但存在核心差异；并以英国地质学一个半世纪的发展为实例，具体阐释了专业化与职业化带来的科研评价标准、期刊发表、学术社群的分化过程；最终总结指出，19世纪是科学结构发生这一核心变革的关键时期。

英文主旨

This article focuses on two major development trends in the scientific field: **specialisation** and **professionalisation**. It opens with the point that specialisation is a product responding to the continuous accumulation of scientific knowledge, and points out that professionalisation is another important reform in the scientific field. Then it analyses that the development of specialisation in the 19th century raised the threshold for amateur participants in scientific research, and there is no absolute boundary between professionals and amateurs but core differences. Taking the one-and-a-half-century development of British geology as an example, it specifically explains the differentiation process of scientific research evaluation standards, journal publication and academic communities brought by specialisation and professionalisation. Finally, it concludes that the 19th century is the crucial period for this core change in the structure of science.

二、逐句精读解析（分段落）

第一段

1. **原文**: Specialisation can be seen as a response to the problem of an increasing accumulation of scientific knowledge.

译文: 专业化可被视为应对科学知识不断积累这一问题的举措。

常考单词&短语:

- specialisation n. 专业化（考研核心词，变体specialization）
- be seen as 被视为；被看作（高频固定搭配）
- a response to 对.....的回应/应对
- accumulation n. 积累；堆积（动词形式accumulate，考研核心词）
- scientific knowledge 科学知识

2. **原文**: By splitting up the subject matter into smaller units, one man could continue to handle the information and use it as the basis for further research.

译文: 通过将研究主题拆分为更小的单元，人们得以继续处理这些信息，并将其作为进一步研究的基础。

常考单词&短语:

- split up 拆分；分割（考研高频固定搭配）
- subject matter 主题；研究对象
- unit n. 单元；单位
- handle v. 处理；应对（核心多义词，此处等同于deal with）
- basis n. 基础；基准（高频搭配on the basis of 基于）
- further research 进一步研究

3. **原文**: But specialisation was only one of a series of related developments in science affecting the process of communication.

译文: 但专业化只是科学领域中一系列影响学术交流进程的相关发展成果之一。

常考单词&短语:

- a series of 一系列的 (高频固定搭配)
- related adj. 相关的; 有关联的
- affect v. 影响; 使改变 (考研核心词, 注意区分effect n. 影响)
- communication n. 交流; 沟通; 传播 (此处为熟词生义, 指学术交流)

长难句结构分析:

- 句子主干: specialisation was only one of a series of related developments
 - 后置定语1: in science 介词短语, 修饰 developments, 限定范围
 - 后置定语2: affecting the process of communication 现在分词短语, 等同于定语从句 which affect the process of communication, 同样修饰 developments, 说明发展成果带来的影响
4. **原文**: Another was the growing professionalisation of scientific activity.
- 译文**: 另一项成果便是科研活动的日益职业化。
- 常考单词&短语**:
- professionalisation n. 职业化 (考研核心词, 变体professionalization, 形容词professional, 名词profession)
 - scientific activity 科研活动; 科学活动

本段段落概括

本段开篇点明专业化的诞生背景——应对科学知识的持续积累, 同时指出专业化并非科学领域唯一的变革, 科研活动的职业化是另一项重要发展趋势, 二者均对学术交流进程产生影响, 为全文的核心论述奠定基础。

第二段

1. **原文**: No clear-cut distinction can be drawn between professionals and amateurs in science: exceptions can be found to any rule.

译文: 在科学领域, 专业人士与业余爱好者之间无法划出泾渭分明的界限: 任何规则都存在例外。

常考单词&短语:

- clear-cut adj. 清晰的; 界限分明的
 - distinction n. 差别; 区分; 界限 (高频搭配draw a distinction between A and B 区分A和B)
 - professional n. 专业人士; adj. 专业的
 - amateur n. 业余爱好者; 外行 (考研核心词)
 - exception n. 例外 (高频搭配exception to the rule 规则的例外)
2. **原文**: Nevertheless, the word “amateur” does carry a connotation that the person concerned is not fully integrated into the scientific community and, in particular, may not fully share its values.
- 译文**: 尽管如此, “业余爱好者”一词确实带有这样的内涵: 相关人员并未完全融入科学界, 尤其可能无法完全认同科学界的价值观念。

常考单词&短语:

- nevertheless adv. 然而; 尽管如此 (考研高频转折逻辑词)
- carry a connotation 带有.....的内涵; 隐含.....的意义
- concerned adj. 相关的; 有关的 (此处作后置定语修饰the person)
- be integrated into 融入; 使并入
- scientific community 科学界; 科研社群
- in particular 尤其; 特别 (高频固定搭配)
- share v. 认同; 分享 (此处为熟词生义, 指“认同(价值观)”)

长难句结构分析:

- 句子主干: the word “amateur” does carry a connotation

- 强调结构: **does carry** 用助动词does对谓语动词carry进行强调, 表“确实、的确”
- 同位语从句: **that the person concerned is not fully integrated into the scientific community and, in particular, may not fully share its values**, 完整解释connotation的具体内容; 从句由and连接两个并列谓语 **is not fully integrated** 和 **may not fully share**, **in particular** 为插入语, 强化递进逻辑。

3. **原文**: The growth of specialisation in the nineteenth century, with its consequent requirement of a longer, more complex training, implied greater problems for amateur participation in science.

译文: 19世纪专业化的发展, 随之带来了对更长期、更复杂的培训的要求, 这给业余爱好者参与科学研究带来了更大的难题。

常考单词&短语:

- nineteenth century 19世纪
- consequent adj. 随之发生的; 作为结果的 (副词consequently, 高频因果逻辑词)
- requirement n. 要求; 必要条件 (动词require, 考研核心词)
- complex adj. 复杂的; 难懂的
- training n. 培训; 训练
- imply v. 意味着; 暗示; 表明 (考研核心词)
- participation n. 参与; 参加 (动词participate, 高频搭配participate in 参与)

长难句结构分析:

- 句子主干: **The growth of specialisation implied greater problems**
- 后置定语: **in the nineteenth century** 介词短语, 修饰主语 **The growth of specialisation**
- 插入语: **with its consequent requirement of a longer, more complex training** 介词短语, 补充说明专业化发展带来的伴随结果
- 后置定语: **for amateur participation in science** 介词短语, 修饰 **problems**, 明确问题针对的对象

4. **原文**: The trend was naturally most obvious in those areas of science based especially on a mathematical or laboratory training, and can be illustrated in terms of the development of geology in the United Kingdom.

译文: 很自然, 这一趋势在那些尤其以数学或实验室培训为基础的科学领域表现得最为明显, 英国地质学的发展便可以佐证这一点。

常考单词&短语:

- trend n. 趋势; 趋向 (考研核心词)
- obvious adj. 明显的; 显而易见的
- be based on 以.....为基础; 基于 (高频固定搭配)
- mathematical adj. 数学的; 数学上的
- laboratory n. 实验室 (简写lab)
- illustrate v. 说明; 阐明; 举例说明 (考研核心词)
- in terms of 就.....而言; 以.....为例 (高频多义短语)
- geology n. 地质学
- the United Kingdom 英国; 联合王国

长难句结构分析:

- 句子主干: **The trend was most obvious... and can be illustrated...**, 由and连接两个并列谓语结构, 共用主语The trend
- 状语与定语: **naturally** 副词作状语修饰was obvious; **in those areas of science** 介词短语作地点状语; **based especially on a mathematical or laboratory training** 过去分词短语作后置定语, 等同于定语从句 **which are based especially on...**, 修饰 **areas of science**; **in terms of the development of geology in the United Kingdom** 介词短语作方式状语, 说明例证的方式。

本段段落概括

本段先指出科学领域的专业人士与业余爱好者并无绝对清晰的界限，但“业余”一词本身带有未完全融入科学界、不认同其核心价值的内涵；随后点明19世纪专业化发展带来的更高培训要求，大幅提升了业余爱好者参与科研的门槛，这一趋势在以数学和实验为基础的学科中尤为突出，并自然引出英国地质学的实例，为下文的具体论证做铺垫。

第三段

1. **原文**: A comparison of British geological publications over the last century and a half reveals not simply an increasing emphasis on the primacy of research, but also a changing definition of what constitutes an acceptable research paper.

译文: 对过去一个半世纪英国地质学出版物的对比研究显示，科学界不仅越来越强调研究的首要地位，同时对什么是合格的科研论文的定义也在不断变化。

常考单词&短语:

- comparison n. 对比；比较（动词compare，高频搭配compare A with B 对比A和B）
- publication n. 出版物；出版；发表（动词publish，考研核心词）
- over the last century and a half 在过去的一个半世纪里
- reveal v. 揭示；显示；透露（考研核心词）
- not simply...but also... 不仅.....而且.....（等同于not only...but also..., 高频并列结构）
- emphasis n. 强调；重视（高频搭配put emphasis on 重视、强调）
- primacy n. 首要地位；至高无上
- definition n. 定义；释义（动词define，考研核心词）
- constitute v. 构成；被视为（考研核心多义词）
- acceptable adj. 可接受的；合格的
- research paper 科研论文；研究论文

长难句结构分析:

- 句子主干: A comparison reveals not simply an increasing emphasis..., but also a changing definition...
 - 后置定语: of British geological publications over the last century and a half 介词短语，修饰主语A comparison
 - 宾语从句: what constitutes an acceptable research paper 作介词of的宾语，整体构成介词短语作后置定语，修饰definition；其中what在宾语从句中充当主语。
2. **原文**: Thus, in the nineteenth century, local geological studies represented worthwhile research in their own right; but, in the twentieth century, local studies have increasingly become acceptable to professionals only if they incorporate, and reflect on, the wider geological picture.

译文: 因此，在19世纪，区域性地质学研究本身就称得上是有价值的科研工作；但到了20世纪，区域性研究只有在融入并深入思考了更宏观的地质学格局时，才会越来越被专业人士所认可。

常考单词&短语:

- thus adv. 因此；从而（高频因果逻辑词）
- local geological studies 区域性地质学研究
- represent v. 代表；相当于（考研核心多义词）
- worthwhile adj. 有价值的；值得做的
- in one's own right 凭自身资格；本身
- increasingly adv. 越来越多地；日益
- only if 只有；只要（引导条件状语从句，表唯一条件，区别于if only 要是.....就好了）
- incorporate v. 包含；融入；合并（考研核心词）
- reflect on 思考；反思；斟酌（高频固定搭配）
- wider geological picture 更宏观的地质学格局；更广阔的地质全貌

长难句结构分析:

- 本句为分号+but连接的并列复合句，前后两个分句形成19世纪与20世纪的时间与逻辑转折对比。

- 前一分句主干: local geological studies represented worthwhile research; in the nineteenth century 为时间状语, in their own right 为方式状语。
- 后一分句主干: local studies have increasingly become acceptable to professionals; only if 引导条件状语从句, 从句的并列谓语为 incorporate, and reflect on, 共同的宾语是 the wider geological picture, 两个谓语动词用逗号隔开作插入语, 表并列。

3. **原文:** Amateurs, on the other hand, have continued to pursue local studies in the old way.

译文: 另一方面, 业余爱好者依然在以传统的方式开展区域性研究。

常考单词&短语:

- on the other hand 另一方面 (高频转折对比逻辑短语)
- pursue v. 从事; 开展; 追求 (此处为熟词生义, 指“开展(研究)”, 考研核心词)
- in the old way 以传统的方式; 用旧方法

4. **原文:** The overall result has been to make entrance to professional geological journals harder for amateurs, a result that has been reinforced by the widespread introduction of refereeing, first by national journals in the nineteenth century and then by several local geological journals in the twentieth century.

译文: 这样带来的总体结果是, 业余爱好者更难在专业地质学期刊上发表文章, 而评审制度的广泛推行进一步加剧了这一结果——评审制度首先在19世纪由全国性期刊引入, 之后在20世纪被多家地方性地质学期刊采用。

常考单词&短语:

- overall adj. 总体的; 全部的
- entrance n. 准入; 进入; 入口 (此处为熟词生义, 指“期刊发表的准入资格”)
- journal n. 期刊; 杂志 (学术场景核心词)
- reinforce v. 加强; 强化; 加剧 (考研核心词)
- widespread adj. 广泛的; 普遍的
- introduction n. 引入; 推行; 介绍 (动词introduce)
- refereeing n. 评审制度; 审稿制度 (动词referee 评审、审稿)
- national journal 全国性期刊

长难句结构分析:

- 句子主干: The overall result has been to make entrance to professional geological journals harder for amateurs, 核心结构为 make sth harder for sb, 意为“让某人做某事更困难”。
- 同位语结构: a result that... 是前面整个主句内容的同位语, 对主句的结果进行补充说明与强化。
- 定语从句: that has been reinforced by the widespread introduction of refereeing 修饰先行词 a result; 从句后接 first by...and then by... 两个并列的方式状语, 清晰说明评审制度引入的时间与主体。

5. **原文:** As a logical consequence of this development, separate journals have now appeared aimed mainly towards either professional or amateur readership.

译文: 这一发展带来的必然结果是, 如今出现了分别主要面向专业读者和业余爱好者读者群体的不同期刊。

常考单词&短语:

- as a logical consequence of 作为.....的必然结果; 因此 (高频因果逻辑短语)
- separate adj. 分开的; 不同的; 各自的
- aim towards 面向; 针对
- either...or... 或者.....或者..... (高频并列结构)
- readership n. 读者群体; 读者人数

长难句结构分析:

- 句子主干: separate journals have now appeared
- 原因状语: As a logical consequence of this development 介词短语, 说明事件的因果逻辑
- 后置定语: aimed mainly towards either professional or amateur readership 过去分词短语, 等同于定语从句 which are aimed mainly towards..., 修饰 separate journals, 说明期刊的定位。

6. **原文**: A rather similar process of differentiation has led to professional geologists coming together nationally within one or two specific societies, whereas the amateurs have tended either to remain in local societies or to come together nationally in a different way.

译文: 一个极为相似的分化过程, 使得专业地质学家在全国范围内聚集到一两个特定的学会之中, 而业余爱好者则往往要么留在地方性学会, 要么以另一种方式在全国范围内聚集。

常考单词&短语:

- rather adv. 相当; 非常
- differentiation n. 分化; 区别 (动词differentiate, 形容词different)
- lead to 导致; 造成 (高频固定搭配, to为介词, 后接名词/动名词)
- geologist n. 地质学家
- come together 聚集; 联合
- specific adj. 特定的; 具体的 (考研核心词)
- society n. 学会; 协会; 社会 (此处为熟词生义, 指“学术学会”)
- whereas conj. 然而; 但是 (高频对比转折连词, 考研完形核心考点)
- tend to do 往往会; 倾向于做某事
- remain v. 保持; 停留; 依然

长难句结构分析:

- 本句为whereas连接的并列复合句, 前后两个分句形成专业人士与业余爱好者的强烈对比, 是考研阅读高频句式。
- 前一分句主干: A rather similar process of differentiation has led to professional geologists coming together...; lead to后接动名词的复合结构 professional geologists coming together..., 其中 professional geologists 是动名词 coming 的逻辑主语。
- 后一分句主干: the amateurs have tended either to remain...or to come together...; 由 either...or... 连接两个并列的不定式结构, 作 tended 的宾语, 清晰说明业余爱好者的两种选择。

本段段落概括

本段以英国地质学一个半世纪的发展为核心实例, 具体论证了专业化与职业化带来的行业变革: 科研论文的评判标准持续提高, 区域性研究的价值判定发生了本质变化; 业余研究者因研究方式未同步更新, 越来越难在专业期刊发表文章, 而评审制度的广泛推行进一步抬高了行业门槛; 最终催生了分别面向专业与业余读者的两类期刊, 以及专业与业余群体彻底分化的学术学会格局, 完整印证了前文的核心观点。

第四段

1. **原文**: Although the process of professionalisation and specialisation was already well under way in British geology during the nineteenth century, its full consequences were thus delayed until the twentieth century.

译文: 尽管19世纪英国地质学领域的职业化与专业化进程早已如火如荼地展开, 但其全部影响直到20世纪才完全显现。

常考单词&短语:

- although conj. 尽管; 虽然 (引导让步状语从句, 高频逻辑词)
- process n. 过程; 进程 (考研核心词)
- be under way 正在进行中; 已启动并推进
- during prep. 在.....期间
- consequence n. 结果; 影响; 后果 (高频核心词)
- delay v. 延迟; 推迟; 耽搁
- until prep./conj. 直到.....为止

2. **原文**: In science generally, however, the nineteenth century must be reckoned as the crucial period for this change in the structure of science.

译文: 但就整个科学领域而言, 19世纪仍必须被视为科学结构发生这一变革的关键时期。

常考单词&短语:

- generally adv. 总体而言；普遍地
- reckon as 被视为；被看作是（等同于be regarded as，固定搭配）
- crucial adj. 关键的；至关重要的（考研核心词，等同于vital/critical）
- period n. 时期；阶段
- structure n. 结构；架构

长难句结构分析：

- 句子主干：the nineteenth century must be reckoned as the crucial period
- 状语：In science generally 介词短语作范围状语；however 副词作插入语，表转折
- 后置定语：for this change in the structure of science 介词短语修饰 period，其中 in the structure of science 又作后置定语修饰 change，明确变革的核心内容。

本段段落概括

本段为全文的总结段落，先承接上文的地质学实例，指出英国地质学的职业化与专业化进程虽在19世纪已全面启动，但其全部影响直到20世纪才完全显现；随后将视角拉回整个科学领域，点明核心结论——19世纪是科学结构发生专业化与职业化这一重大变革的关键时期，收束全文，升华主旨。

三、完整原文

Specialisation can be seen as a response to the problem of an increasing accumulation of scientific knowledge. By splitting up the subject matter into smaller units, one man could continue to handle the information and use it as the basis for further research. But specialisation was only one of a series of related developments in science affecting the process of communication. Another was the growing professionalisation of scientific activity.

No clear-cut distinction can be drawn between professionals and amateurs in science: exceptions can be found to any rule. Nevertheless, the word "amateur" does carry a connotation that the person concerned is not fully integrated into the scientific community and, in particular, may not fully share its values. The growth of specialisation in the nineteenth century, with its consequent requirement of a longer, more complex training, implied greater problems for amateur participation in science. The trend was naturally most obvious in those areas of science based especially on a mathematical or laboratory training, and can be illustrated in terms of the development of geology in the United Kingdom.

A comparison of British geological publications over the last century and a half reveals not simply an increasing emphasis on the primacy of research, but also a changing definition of what constitutes an acceptable research paper. Thus, in the nineteenth century, local geological studies represented worthwhile research in their own right; but, in the twentieth century, local studies have increasingly become acceptable to professionals only if they incorporate, and reflect on, the wider geological picture. Amateurs, on the other hand, have continued to pursue local studies in the old way. The overall result has been to make entrance to professional geological journals harder for amateurs, a result that has been reinforced by the widespread introduction of refereeing, first by national journals in the nineteenth century and then by several local geological journals in the twentieth century. As a logical consequence of this development, separate journals have now appeared aimed mainly towards either professional or amateur readership. A rather similar process of differentiation has led to professional geologists coming together nationally within one or two specific societies, whereas the amateurs have tended either to remain in local societies or to come together nationally in a different way.

Although the process of professionalisation and specialisation was already well under way in British geology during the nineteenth century, its full consequences were thus delayed until the twentieth century. In science generally, however, the nineteenth century must be reckoned as the crucial period for this change in the structure of science.

四、完整全文译文

专业化可被视为应对科学知识不断积累这一问题的举措。通过将研究主题拆分为更小的单元，人们得以继续处理这些信息，并将其作为进一步研究的基础。但专业化只是科学领域中一系列影响学术交流进程的相关发展成果之一。另一项成果便是科研活动的日益职业化。

在科学领域，专业人士与业余爱好者之间无法划出泾渭分明的界限：任何规则都存在例外。尽管如此，“业余爱好者”一词确实带有这样的内涵：相关人员并未完全融入科学界，尤其可能无法完全认同科学界的价值观念。19世纪专业化的发展，随之带来了对更长期、更复杂的培训的要求，这给业余爱好者参与科学研究带来了更大的难题。很自然，这一趋势在那些尤其以数学或实验室培训为基础的科学领域表现得最为明显，英国地质学的发展便可以佐证这一点。

对过去一个半世纪英国地质学出版物的对比研究显示，科学界不仅越来越强调研究的首要地位，同时对什么是合格的科研论文的定义也在不断变化。因此，在19世纪，区域性地质学研究本身就称得上是有价值的科研工作；但到了20世纪，区域性研究只有在融入并深入思考了更宏观的地质学格局时，才会越来越被专业人士所认可。另一方面，业余爱好者依然在以传统的方式开展区域性研究。这样带来的总体结果是，业余爱好者更难在专业地质学期刊上发表文章，而评审制度的广泛推行进一步加剧了这一结果——评审制度首先在19世纪由全国性期刊引入，之后在20世纪被多家地方性地质学期刊采用。这一发展带来的必然结果是，如今出现了分别主要面向专业读者和业余爱好者读者群体的不同期刊。一个极为相似的分化过程，使得专业地质学家在全国范围内聚集到一两个特定的学会之中，而业余爱好者则往往要么留在地方性学会，要么以另一种方式在全国范围内聚集。

尽管19世纪英国地质学领域的职业化与专业化进程早已如火如荼地展开，但其全部影响直到20世纪才完全显现。但就整个科学领域而言，19世纪仍必须被视为科学结构发生这一变革的关键时期。